INTERNET OF THINGS

TASK--2

SENSOR – BASED IOT SIMULATION USING ARDUINO AND LDR

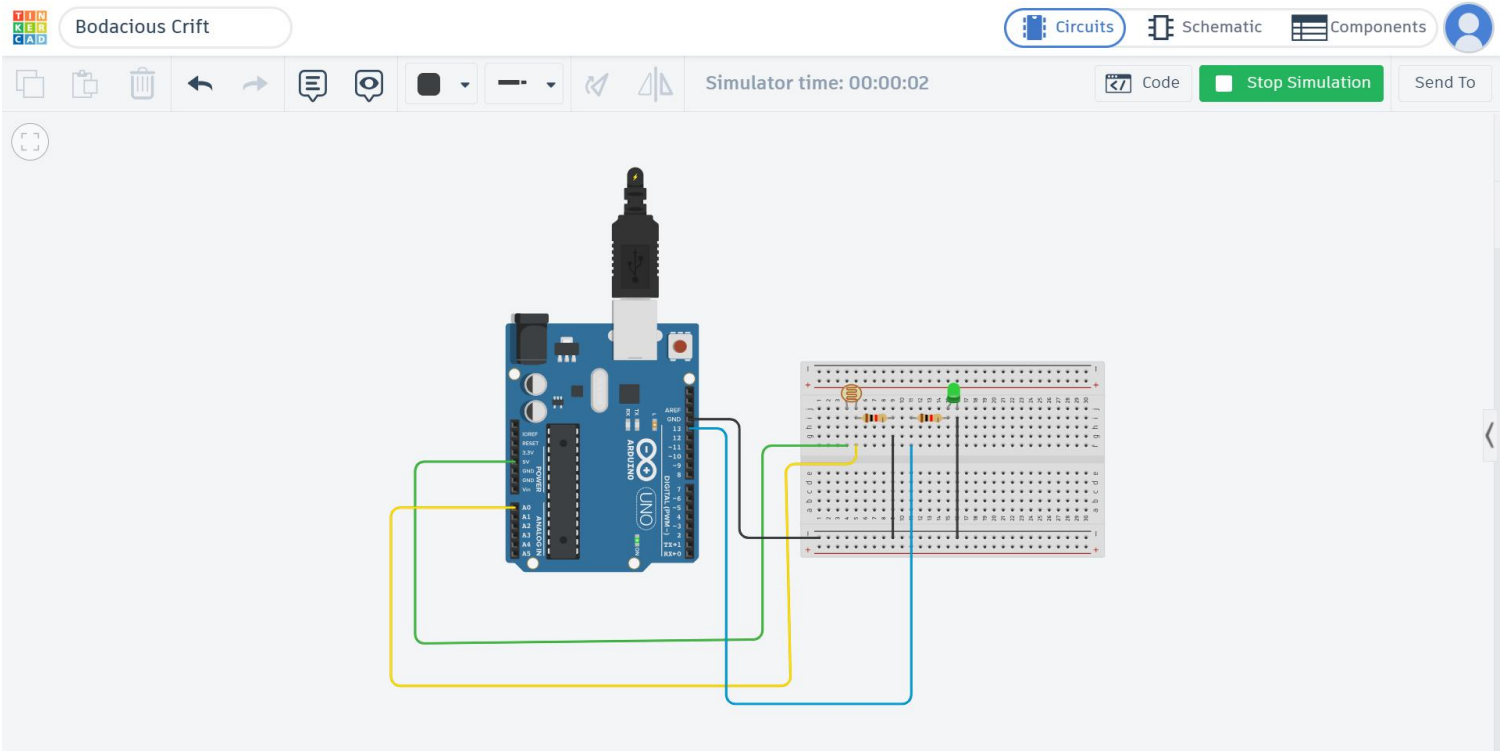
AIM:- To simulate a light-based automatic LED control system using Arduino Uno and an LDR sensor in Tinkercad.

Components Required :-

- Arduino Uno
- Breadboard
- LDR (Light Dependent Resistor)
- 10 kΩ resistor
- LED- 220 Ω resistor
- Connecting wires

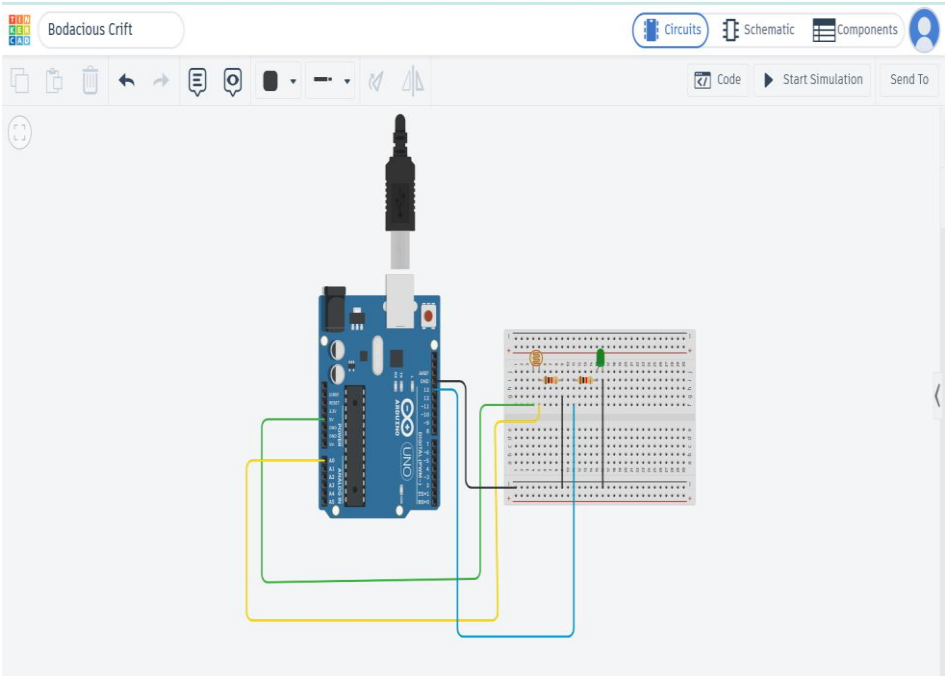
Working Principle :- The LDR sensor detects the light intensity. Arduino reads the sensor value and controls the LED. When the light level is below 500, the LED turns ON. Otherwise, the LED turns OFF.

Circuit Diagram :-

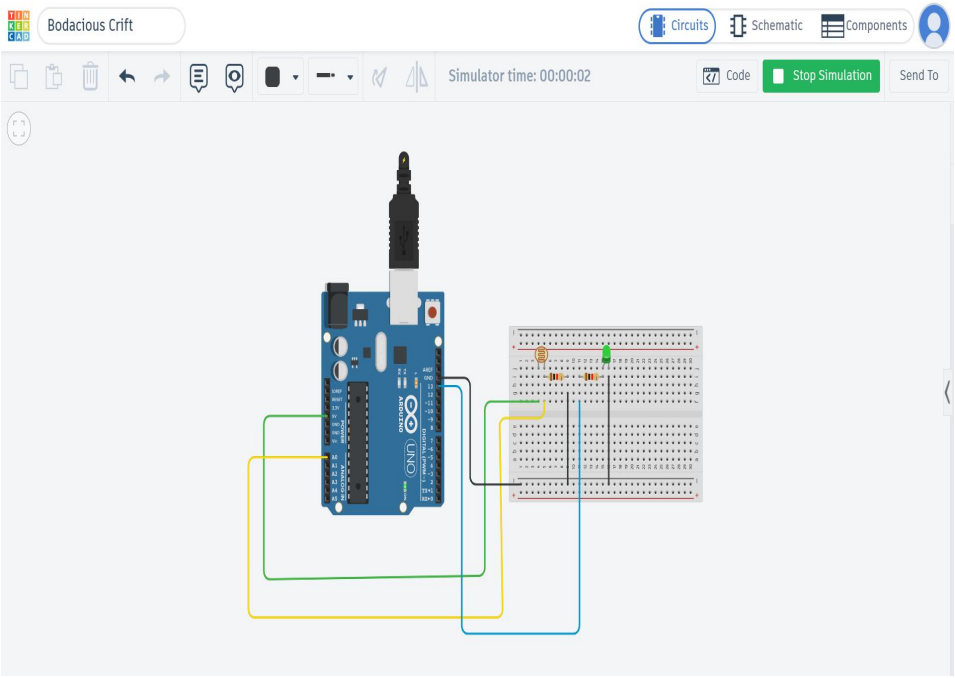


Simulation Output :-

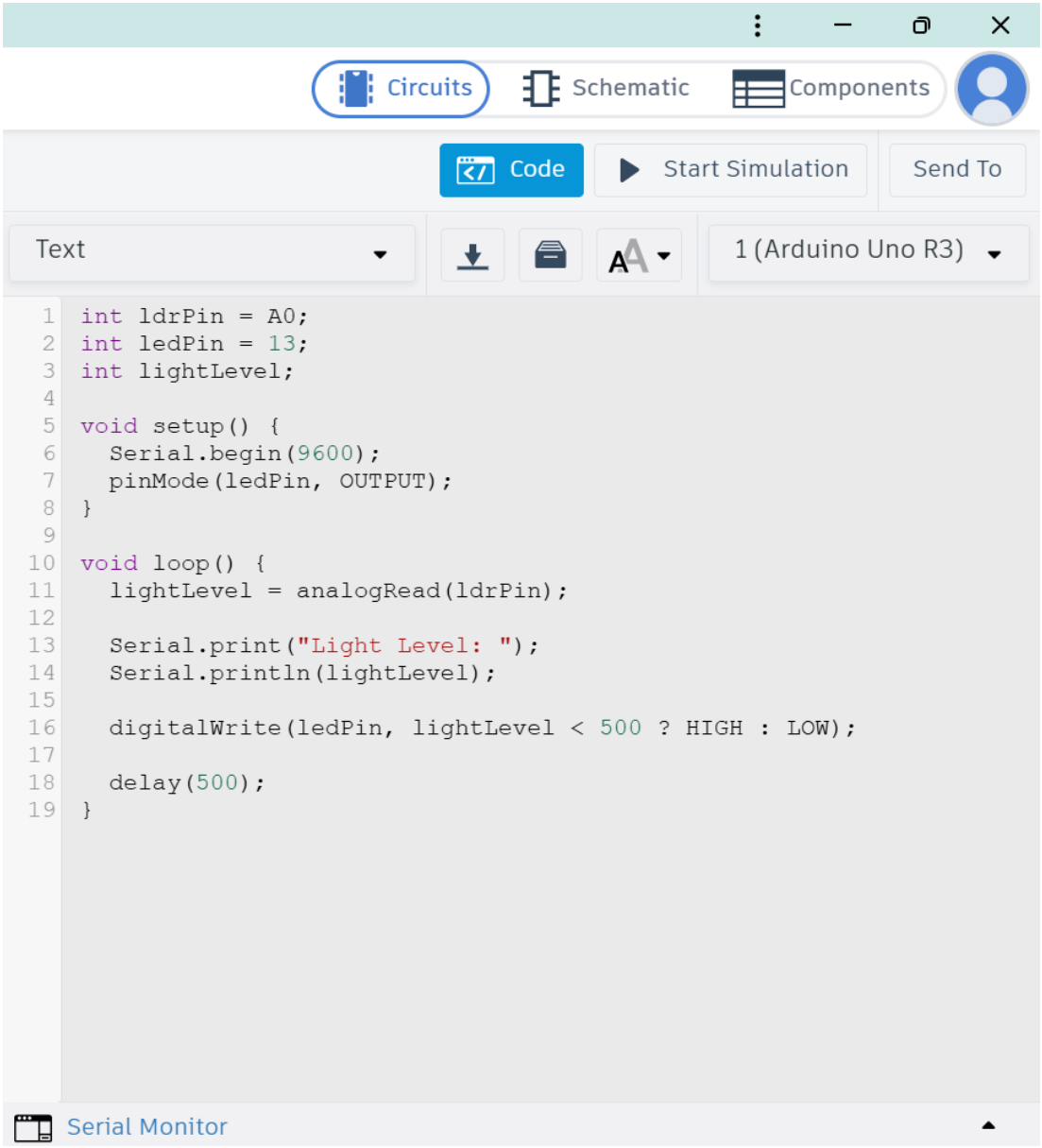
LED OFF



LED ON



Arduino Code :-




```
int ldrPin = A0;
int ledPin = 13;
int lightLevel;

void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
}

void loop() {
  lightLevel = analogRead(ldrPin);

  Serial.print("Light Level: ");
  Serial.println(lightLevel);

  digitalWrite(ledPin, lightLevel < 500 ? HIGH : LOW);

  delay(500);
}
```

CODE EXPLANATION :-

The Arduino program is used to control an LED based on the light intensity detected by an LDR sensor. The Arduino Uno reads the sensor value and automatically turns the LED ON or OFF according to the given condition.

1. Variable Declaration :-

"int ldrPin = A0;"

This line connects the LDR sensor input to analog pin A0 of the Arduino. The sensor value is read through this pin.

"int ledPin = 13;"

This line assigns digital pin 13 to control the LED.

"int lightLevel;"

This variable stores the light intensity value received from the LDR sensor.

2. Setup Function :-

"void setup()"

The setup function runs only once when the Arduino program starts.

"Serial.begin(9600);"

This statement starts serial communication at a speed of 9600 bits per second. It helps display the sensor readings in the Serial Monitor.

"pinMode(ledPin, OUTPUT);"

This statement sets digital pin 13 as an output pin because the Arduino uses this pin to control the LED.

3. Loop Function

"void loop()“

The loop function runs repeatedly while the simulation is running. It continuously reads the light level and controls the LED.

4. Reading the LDR Sensor

"lightLevel = analogRead(ldrPin);“

This statement reads the analog value from the LDR sensor through pin A0. The value ranges from 0 to 1023.

5. Displaying the Sensor Value

"Serial.print("Light Level: ");“

This statement displays the text “Light Level:” in the Serial Monitor.

"Serial.println(lightLevel);“

This statement displays the current sensor reading and moves the cursor to the next line.

6. Controlling the LED

"digitalWrite(ledPin, lightLevel < 500 ? HIGH : LOW);“

This statement checks the sensor value and controls the LED.

- If the light level value is less than 500, the LED turns ON.
- If the light level value is 500 or greater, the LED turns OFF.The value 500 is the threshold value used to decide when the LED should turn ON or OF

7. Delay Function

"delay(500);“

This statement pauses the program for 500 milliseconds before the next sensor reading. It allows the program to update the readings at regular intervals.

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Overall Working :-

The LDR sensor detects the surrounding light intensity and sends a signal to the Arduino. The Arduino reads the sensor value, compares it with the threshold value of 500, and controls the LED accordingly. The sensor readings are also displayed in the Serial Monitor.This project demonstrates how a sensor and a microcontroller can work together to perform automatic light control.

9. Result :-

The light-based LED control system was simulated using Arduino Uno and an LDR sensor in Tinkercad. The LED is controlled according to the detected light level.